LESSON 1: DRUG TOXICITY AND POISONING

Therapeutic index, types of toxicity, interactions, antidotes

PHARMACOLOGY | MIDTERM: TOXICITY, METABOLISM, AUTONOMICS, ANTIDIABETICS

How to study this lesson

Learn it as **lists you can rebuild**: 4 types of toxicity, 4 hypersensitivity types (ACID), 5 levels of drug interaction, 4 kinds of antagonism, then the **antidote groups**.

THERAPEUTIC INDEX AND MARGIN OF SAFETY

- **Therapeutic index (TI)** = **TD50 ÷ ED50** (toxic dose in 50% ÷ effective dose in 50%)
  - Example from the slide: TD50 6 ÷ ED50 2 = **TI of 3**
  - **Higher TI = safer drug**
- **Margin of safety** = **LD1 ÷ ED99** (lethal dose in 1% ÷ effective dose in 99%)
  - Fixes the weakness of TI: uses the **extremes** of the curves
  - Slide example: Drug A = 10/10 = **1**; Drug B = 0.002/10 = **0.0002**, so Drug B is **much less safe**

Note: Memory tip: TI uses the **middles** (the 50s); margin of safety uses the **edges** (1 and 99). "Toxic over effective", always the bad number on top.

TYPES OF THERAPEUTIC DRUG TOXICITY

- Drug effects are **desirable (therapeutic)** or **undesirable**
  - Undesirable = **nondeleterious (side effects)** or **deleterious (toxic effects)**
  - Toxic effects = **pharmacological**, **pathological** or **genotoxic**
- Four types to know: **Dose-dependent**, **Allergic**, **Idiosyncratic**, **Drug-drug interactions**

Note: Memory tip: "**DAID**" – **D**ose, **A**llergy, **I**diosyncrasy, **D**rug-drug.

DOSE-DEPENDENT REACTIONS

- Severity rises with **drug concentration** and **duration of exposure**
- **Pharmacological toxicity**
  - **CNS depression by barbiturates** – **dose**-dependent
  - **Tardive dyskinesia from antipsychotics** – **duration**-dependent
  - **Phototoxicity** – **tetracyclines**
- **Extrapyramidal symptoms** (antipsychotics), in order of onset:
  - **Acute dystonia** – hours to 5 days; spasm of tongue, neck, face, back
  - **Parkinsonism** – 5–30 days; tremor, shuffling gait, drooling, stooped posture
  - **Akathisia** – 5–60 days; compulsive repetitive motion, restlessness
  - **Tardive dyskinesia** – months to years; lip-smacking, worm-like tongue, "fly-catching"

Note: Memory tip: EPS go **D-P-A-T** in time order: "**D**on't **P**anic, **A**ntipsychotics **T**ake time". Tardive = **tardy** (late).

- **Pathological toxicity** – e.g. **acetaminophen**: CYP2E1 makes **NAPQI**, which causes **hepatocyte necrosis** when glutathione runs out
- **Genotoxic effects** – **ionizing radiation, environmental chemicals, chemotherapy** injure DNA → **mutagenic or carcinogenic** toxicity

ALLERGIC REACTIONS (HYPERSENSITIVITY)

- **Type I – Anaphylactic** – IgE on mast cells and basophils → **histamine** release (bronchoconstriction, vasodilation, capillary leak, swelling)
- **Type II – Cytolytic** (cytotoxic)
- **Type III – Arthus reaction** (immune complex)
- **Type IV – Delayed** (T-cell mediated)

Note: Memory tip: "**ACID**" – **A**naphylactic, **C**ytotoxic, **I**mmune complex, **D**elayed. Type I is the one with **IgE + histamine**.

- **Antibodies:** IgE = **allergy** and parasites (binds mast cells, basophils); IgG = main blood antibody, **crosses the placenta**; IgM = first responder, fixes complement; IgA = secretions (saliva, tears, colostrum); IgD = B-cell receptor

IDIOSYNCRATIC REACTIONS

- **Idiosyncrasy** – abnormal reactivity to a chemical that is **peculiar to an individual** (usually genetic)
- Example: **Primaquine** (antimalarial) causes **hemolytic anemia in 10% of black males**

DRUG-DRUG INTERACTIONS

- Five levels: **Absorption** → **Protein binding** → **Metabolism** → **Receptor binding** → **Therapeutic action**

Note: Memory tip: follow the drug's journey: it is **absorbed**, rides on **proteins**, gets **metabolized**, hits a **receptor**, then makes an **effect**.

- **Absorption** – **Ranitidine** (H2 blocker) **raises GI pH** → may **increase absorption of basic drugs** like **triazolam**
- **Protein binding** – highly bound drugs: **aspirin, barbiturates, phenytoin, sulfonamides, valproic acid, warfarin**
  - A second drug with higher affinity **displaces** the first → more **free (active) drug**
- **Metabolism** – usually via **hepatic CYPs**; **ethanol is a potent inducer of CYP2E1**
- **Receptor binding** – **Buprenorphine**: opioid **partial agonist/antagonist**; treats opioid addiction and **blocks the euphoria** of other narcotics taken with it
- **Therapeutic action** – **additive** (1 + 1 = 2), **synergistic** (1 + 1 > 2), **antagonistic** (1 + 1 < 2)

KINDS OF ANTAGONISM

- **Functional / physiological** – two drugs act on **different receptors** with **opposite effects**
- **Chemical** – one drug **binds and inactivates** the other (e.g. **chelators**: **EDTA** binds Ca2+ and lead into water-soluble complexes)
- **Dispositional** – one drug changes the other's **absorption, distribution, metabolism or excretion**
- **Receptor** – **competitive** (surmountable, same site) vs **noncompetitive** (unsurmountable)

Note: Memory tip: "**F-C-D-R**": **F**ight from another receptor, **C**hemically grab it, **D**ispose of it, or block the **R**eceptor.

PREVENTION AND TREATMENT OF POISONING

- **Medication errors** are **50–100× more common** than adverse drug reactions
- Home prevention: **passive** (no behavior change needed, e.g. child-proof caps) vs **active** (needs sustained adaptation)
- Reducing medication errors: unit-dose systems, pharmacy-prepared IV solutions, remove dangerous drugs (concentrated KCl) from wards; long term: **computerized order entry**, dose and allergy checking, **bar codes**

TREATMENT PRIORITIES

- 1) **Maintain vital functions** 2) **Reduce absorption and enhance elimination** 3) **Combat the poison's effects** at its site of action
- Initial stabilization – **ABCDE**: **A**irway, **B**reathing, **C**irculation, **D**isability (CNS; consider O2, dextrose, naloxone, thiamine), **E**xposure ("toxidrome")

DECONTAMINATION

- **First step: stop ongoing exposure**; wash eyes and skin copiously
- GI decontamination only if: the poison is **potentially dangerous**, **still unabsorbed** in the stomach or intestine, and the procedure can be done **safely**

- Methods: **activated charcoal** (binds the toxicant in the gut), **whole-bowel irrigation**, **cathartics**, **gastric lavage**, **syrup of ipecac**

ANTIDOTES (GROUPED SO THEY'RE EASIER TO REMEMBER)

METALS: CHELATORS

- **Dimercaprol (BAL)** – lead, mercury, arsenic
- **Succimer (DMSA)** – lead, mercury, arsenic
- **EDTA (CaNa2)** – **lead**
- **Penicillamine** – lead, mercury, **copper**
- **Deferoxamine** – **iron**

Note: Memory tip: "**DE-FE-roxamine**" has **Fe** (iron) in it. **Peni-copper**: penicillamine is the copper one.

TOXIC ALCOHOLS

- **Fomepizole** or **ethanol** – **methanol, ethylene glycol** (both block alcohol dehydrogenase)

REVERSE A RECEPTOR DRUG

- **Naloxone** – **opioids**
- **Flumazenil** – **benzodiazepines**
- **Atropine** + **pralidoxime (2-PAM)** – **organophosphate** (and carbamate) pesticides
- **Physostigmine** – **anticholinergic syndrome**
- **Glucagon** – **β-blockers**
- **High-dose insulin** and **calcium** – **Ca2+ channel blockers**
- **Digoxin immune Fab** – cardiac glycosides (digoxin)
- **Sodium bicarbonate** – Na+ channel-blocking drugs

BLOOD AND METABOLISM

- **Hydroxocobalamin** or **sodium thiosulfate** – **cyanide**
- **Methylene blue** – **methemoglobinemia**
- **Hyperbaric oxygen** – **carbon monoxide**
- **Vitamin K1 (phytonadione)** – coumarin (warfarin), indandione
- **Leucovorin** – **methotrexate**
- **Octreotide** – **sulfonylurea-induced hypoglycemia**

DRUG-SPECIFIC

- **N-acetylcysteine (NAC)** – **acetaminophen** (refills glutathione)
- **Pyridoxine (B6)** – **isoniazid seizures**
- **Carnitine** – valproate hyperammonemia
- **Benztropine** or **diphenhydramine** – drug-induced **dystonia**
- **Bromocriptine** – **neuroleptic malignant syndrome**
- **Dantrolene** – **malignant hyperthermia**
- **Crotalidae immune Fab** – rattlesnake (crotaline) envenomation

Note: Memory tip: match the antidote to the **problem's mechanism**: metals get **chelated**, receptor drugs get a **blocker or reverser**, and poisons that use up a body substance get it **replaced** (NAC replaces glutathione, B6 replaces pyridoxine, glucagon bypasses blocked β-receptors).

———— END OF LESSON ————

LESSON 2: DRUG METABOLISM

Phase 1 and 2 reactions, CYPs, conjugation enzymes

PHARMACOLOGY | MIDTERM: TOXICITY, METABOLISM, AUTONOMICS, ANTIDIABETICS

What is drug metabolism for?

Turning **xenobiotics** (substances foreign to the body) into forms the body can **eliminate**. Drugs reuse the same enzymes and transporters the body uses for dietary substances.

- Liver sequence: **Phase I (oxidation, CYP450)** → **Phase II (conjugation)** → **Phase III (transport, ABC transporters)** → excretion in **urine or bile**

PHASE 1 VS PHASE 2

- **Phase 1** – **oxidation, reduction, hydrolysis**
  - Adds or exposes a **functional group** (–OH, –COOH, –SH, –O–, NH2)
  - **Little increase in water solubility**, but **dramatically changes** the drug's activity (often **inactivation**)
- **Phase 2** – **conjugation**
  - Attaches a large polar group → **much more water-soluble** metabolite → **easier elimination**
- Example: **phenytoin** (highly lipophilic) → Phase 1 (CYP) → **4-hydroxyphenytoin** (slightly soluble) → Phase 2 (UGT) → **glucuronide** (very soluble)

Note: Memory tip: Phase **1** = "**1** small change" (add a handle). Phase **2** = "**2**gether": **attach** a big group to the handle so it washes out.

PHASE 1 ENZYMES

CYPS (CYTOCHROME P450 SUPERFAMILY)

- **Heme** bound noncovalently to a polypeptide (hemoproteins); handle about **75% of drug metabolism**
- Reactions: **N- and O-dealkylation, aromatic hydroxylation, N- and S-oxidation, deamination, dehalogenation**
- Also make **steroids and bile acids**; **CYP19 (aromatase)** is inhibited by **anastrozole**
- **Naming: CYP3A4** = family **3**, subfamily **A**, gene **4**
- Families **1, 2 and 3** do drug metabolism; most active: **CYP2C, CYP2D, CYP3A**
- **CYP3A4 metabolizes over 50% of drugs**

FMOS (FLAVIN-CONTAINING MONOOXYGENASES)

- **Minor** contributor; **benign products**; **fewer drug-drug interactions**
- Six families; **FMO3** is the most abundant
- Metabolize: **nicotine, H2 blockers (cimetidine, ranitidine), clozapine, itopride**

OTHER PHASE 1 ENZYMES

- **Epoxide hydrolases** (hydrolysis of epoxides); **alcohol and aldehyde dehydrogenases**; **NQO**

CYP INDUCERS AND INHIBITORS

- **Inducers** speed up metabolism → **lower** levels of other drugs (less effect)
- **Inhibitors** slow metabolism → **higher** levels of other drugs (toxicity)
- **Inducers – "BULLSHIT CRAP GP'S"** (slides): **B**arbiturates, **S**t. John's Wort, **C**arbamazepine, **R**ifampin, **A**lcohol (**chronic**), **P**henytoin, **G**riseofulvin, **P**henobarbital, **S**ulfonylureas
- **Inhibitors – "SICKFACES.COM"** (slides): **S**odium valproate, **I**soniazid, **C**imetidine, **K**etoconazole, **F**luconazole, **A**lcohol (**acute** binge), **C**hloramphenicol, **E**rythromycin, **S**ulfonamides, **C**iprofloxacin, **O**meprazole, **M**etronidazole
  - Plus **grapefruit juice, amiodarone, quinidine**

Note: Alcohol appears in **both**: **chronic** drinking **induces**, an **acute** binge **inhibits**. Example: **St. John's Wort induces CYP3A4**, so its substrates are cleared faster.

PHASE 2 ENZYMES ("TRANSFERASES")

- **UGTs (glucuronidation)** – glucuronic acid from **UDP-GA**; the largest share of Phase 2
  - Drugs: **acetaminophen, morphine, oxazepam, lorazepam**
  - **UGT1A1** glucuronidates **bilirubin**; defects cause **Gilbert** and **Crigler-Najjar** syndromes (jaundice); **phenobarbital** induces UGT1A1
- **SULTs (sulfation)** – sulfate from **PAPS**
  - Drugs: **acetaminophen, steroids, methyldopa**
  - **SULT2B1b** (skin) makes **cholesterol sulfate** for keratinocyte differentiation
  - **SULT2A1** (fetal adrenal) makes **DHEA-S** for **placental estrogen** in the 2nd half of pregnancy
- **GSTs (glutathione conjugation)** – **glutathione** = tripeptide of **glutamic acid, cysteine, glycine**
  - Drugs: **adriamycin, fosfomycin, busulfan**
- **NATs (N-acetylation)** – drugs with an **aromatic amine or hydrazine** group
  - Drugs: **sulfonamides, isoniazid, dapsone, clonazepam** (also hydralazine, procainamide)
  - **Acetylator phenotype** (fast vs slow): slow acetylators get toxicity, e.g. **isoniazid** → **peripheral neuritis, hepatotoxicity**; **procainamide** → **lupus**
- **MTs (methylation)** – O-, N-, S-methylation using **S-adenosylmethionine (SAM)** as methyl donor
  - Drugs: **L-dopa, methyldopa, mercaptopurine, captopril**; **COMT** methylates catecholamines (NE → normetanephrine)
  - **Azathioprine** → **6-mercaptopurine**, cleared by **TPMT** (methylation) and **xanthine oxidase**; low TPMT → toxic buildup of 6-thioguanine nucleotides

Note: Memory tip: "**GAS** on **SAM**": **G**lucuronic acid, **A**cetyl, **S**ulfate, glutathione and methyl from **SAM**. Each enzyme's name is the group it adds: **U**DP-**G**lucuronosyl**T**ransferase adds **glucuronic acid**, **S**ulfo**T**ransferase adds **sulfate**, and so on.

ACETAMINOPHEN: PHASE 1 AND 2 TOGETHER

- About **90%** → **glucuronidation and sulfation** (nontoxic)
- About **5–9%** → **CYP2E1** → **NAPQI** (toxic) → neutralized by **glutathione**
- **Overdose**: glutathione runs out → NAPQI → **hepatocyte necrosis**
- Antidote: **N-acetylcysteine (NAC)** replenishes glutathione

Note: CYP2E1 is the enzyme **ethanol induces**, which is why drinking worsens acetaminophen liver damage.

———— END OF LESSON ————

LESSON 3: AUTONOMICS I

ANS review and cholinergic agonists

PHARMACOLOGY | MIDTERM: TOXICITY, METABOLISM, AUTONOMICS, ANTIDIABETICS

How should I sort autonomic drugs?

Ask 3 questions: **which system** (cholinergic or adrenergic)? **Mimic or block? Direct** (on the receptor) or **indirect** (on the enzyme or nerve terminal)? Every drug in Lessons 3 and 4 fits one box.

THE AUTONOMIC NERVOUS SYSTEM

- Nervous system functions: **sensory input, integration, motor output**
- ANS = involuntary regulation of internal organs
  - **Sympathetic** – **fight or flight**; thoracolumbar (T1–L2); short preganglionic, long postganglionic; ganglia near the spinal cord
  - **Parasympathetic** – **rest and digest**; craniosacral (brainstem, S2–S4); long preganglionic, short postganglionic; ganglia near the organ
- Neurotransmitters:
  - **All preganglionic fibers** → **ACh** (on **nicotinic Nn** receptors)
  - **Parasympathetic postganglionic** → **ACh** (on **muscarinic** receptors)
  - **Sympathetic postganglionic** → **NE** (on **adrenergic** receptors)
  - **Exception:** sympathetic fibers to **eccrine sweat glands use ACh**
  - **Adrenal medulla** releases **epinephrine (and NE)** into the blood
  - **Somatic** NMJ → **ACh on Nm** receptors

Note: Memory tip: "ACh everywhere except sympathetic endings." Sweat glands are the exception's exception.

TOXIDROMES (SLIDES)

- **Anticholinergic** (= sympathetic-like): **Hot as a hare** (hyperthermia), **Dry as a bone** (no secretions), **Red as a beet** (vasodilation), **Blind as a bat** (cycloplegia, mydriasis), **Mad as a hatter** (CNS toxicity)
- **Cholinergic / organophosphate – "DUMBBELLS"**: **D**iarrhea/Diaphoresis, **U**rination, **M**iosis, **B**radycardia, **B**ronchospasm/Bronchorrhea, **E**mesis, **L**ethargy, **L**acrimation, **S**alivation

Note: They are **opposites**: cholinergic = **wet** (everything leaks), anticholinergic = **dry and hot**.

ACETYLCHOLINE LIFE CYCLE (AND WHAT BLOCKS EACH STEP)

- **Transport of choline** – choline transporter (CHT, Na+/choline symport) → blocked by **hemicholinium**
- **Synthesis** – **choline acetyltransferase (ChAT)**: acetyl-CoA + choline → ACh
- **Storage** – vesicles via **VAT** → blocked by **vesamicol**
- **Release** – Ca2+ entry → VAMPs and SNARE proteins → blocked by **botulinum toxin**
- **Binding** – to postsynaptic receptor
- **Termination** – **acetylcholinesterase (AChE)** → choline + acetate → blocked by **indirect cholinomimetics** (carbamates, organophosphates)

Note: Memory tip: "He Vests Botox, then Stops": **He**micholinium (uptake), **Ves**amicol (storage), **Bot**ulinum (release), **-stigmines** (stop AChE).

CHOLINERGIC RECEPTORS

- **Muscarinic** (G-protein): **M1** nerves (IP3/DAG), **M2** heart (↓cAMP, opens K+ channels → slows heart), **M3** glands and smooth muscle (IP3/DAG), **M4, M5** CNS
- **Nicotinic** (ion channels, Na+/K+): **Nm** = skeletal muscle **NMJ**; **Nn** = **ganglia** and CNS

Note: Memory tip: M**2** = heart ("**2** words: heart rate"); M**3** = glands ("**3** secretions: saliva, tears, sweat"). **N-m = muscle**, **N-n = neurons (ganglia)**.

CHOLINERGIC AGONISTS (PARASYMPATHOMIMETICS, CHOLINOMIMETICS)

- **Direct-acting** – bind the receptor: muscarinic (choline esters, alkaloids) or nicotinic
- **Indirect-acting** – **do not bind** cholinoceptors; **inhibit AChE** so ACh lasts longer

DIRECT-ACTING MUSCARINIC

- **Acetylcholine** – activates **both** nicotinic and muscarinic; use: **miotic eye drop**
- **Choline esters** (nonselective M1–M3):
  - **Bethanechol** – **bladder and bowel retention**
  - **Methacholine** – triggers bronchoconstriction to **diagnose asthma**
- **Alkaloid** (selective M3, glands):
  - **Pilocarpine** – lacrimal and salivary secretion; for **Sjögren and sicca syndrome**; SE: **sweating** (most common)
- **Muscarinic toxicity** → treat with **atropine**

Note: Memory tip: Bethanechol for the **bath**room (bladder, bowel); **Metha**choline for **me**asuring asthma; **Pilo**carpine makes you **pile** up spit and tears.

DIRECT-ACTING NICOTINIC

- **Neural and muscular agonists** – **nicotine** (full), **varenicline** (partial) → **smoking cessation**
- **Muscular (Nm) agonist** – **succinylcholine (suxamethonium)** → **depolarizing NMJ blocker**
  - **Phase I:** membrane depolarizes → transient **fasciculations** → **flaccid paralysis**
  - **Phase II:** membrane repolarizes but the receptor is **desensitized** to ACh

INDIRECT-ACTING (ACHE INHIBITORS), BY DURATION

- **Short-acting – edrophonium** (reversible) – **bedside diagnosis of myasthenia gravis**
- **Intermediate-to-long – carbamates (-STIGMINES)**
  - **Neostigmine, pyridostigmine** – **treatment of myasthenia gravis**
  - **Physostigmine** – **enters the CNS** → fixes **atropine overdose** (anticholinergic syndrome)
- **Very long-acting – organophosphates (-THION)**
  - **Parathion, malathion** – **irreversible** AChE inhibition; **pesticides**
  - **Tabun, sarin, soman** – **war nerve gases** (WW2)
- **Drugs for Alzheimer dementia** (AChE inhibitors; cholinergic neurons are deficient in Alzheimer): **rivastigmine, donepezil, galantamine**

Note: Slide mnemonic: "**Si Don Pepe, may dementia. Pa Gala Gala sa Riva**" – **Don**epezil, **Gala**ntamine, **Riva**stigmine.

Note: Memory tip: Edrophonium = **E**xam (diagnosis, very short). **Physo**stigmine = **Physically gets into the brain** (tertiary amine), so it reverses central atropine toxicity. **-THION = THE ONE you can't undo** (irreversible).

———— END OF LESSON ————

LESSON 4: AUTONOMICS II

Anticholinergics and adrenergic drugs

PHARMACOLOGY | MIDTERM: TOXICITY, METABOLISM, AUTONOMICS, ANTIDIABETICS

CHOLINERGIC ANTAGONISTS (PARASYMPATHOLYTICS, ANTICHOLINERGICS)

- Three groups: **antimuscarinic**, **antinicotinic**, **cholinesterase regenerators** (oximes, e.g. **pralidoxime**)

MUSCARINIC BLOCKERS

- **Nonselective** (prototype **atropine**):
  - **Atropine, homatropine, cyclopentolate, tropicamide** – eye (**mydriatics**)
  - **Ipratropium, tiotropium, umeclidinium, glycopyrronium** – airways
  - **Scopolamine**
- **Atropine** – blocks **all** muscarinic receptors; **mydriatic**; **1st choice for organophosphate poisoning**; **symptomatic bradycardia**; with anesthesia for bradycardia, hypersalivation, airway secretions
- **Ipratropium, tiotropium, umeclidinium, glycopyrronium** – block bronchial muscarinic receptors → **acute asthma and COPD**; **tiotropium has the longest half-life**
- **Scopolamine** – stereoisomer of atropine, **crosses the BBB better → motion sickness**
- **M3 blockers – dicyclomine (dicycloverine)** for **diarrhea**; **oxybutynin** for **urge incontinence**
- **M1-selective – pirenzepine, telenzepine** → ↓ gastric acid → **peptic ulcer disease**

Note: Memory tip: "**-tropium**" drugs are **inhaled** for the airways (asthma, COPD). **Scop**olamine = **sea** sickness (motion). **O**xybutynin = **O**veractive bladder. **D**icyclomine = **D**iarrhea.

NICOTINIC BLOCKERS

- **Ganglionic blocker – hexamethonium** (**Nn**); was used for **hypertension** (obsolete)
- **Neuromuscular blockers (curares) – tubocurarine, pancuronium, atracurium, vecuronium** (**Nm**); **skeletal muscle relaxation before surgery**

Note: Memory tip: "**-curonium / -curium = curare = surgery**". Hexa-**METH**onium blocks the **middle man** (the ganglion).

CHOLINERGIC SUMMARY TABLE (FILL-IN FROM THE SLIDES, COMPLETED)

- **Agonists, direct:** nonselective = **bethanechol, methacholine**; M3-selective = **pilocarpine**; Nn and Nm = **nicotine, varenicline**; Nm = **succinylcholine**
- **Agonists, indirect:** shortest = **edrophonium**; intermediate = **neostigmine, pyridostigmine, physostigmine**; organophosphates = **parathion, malathion, nerve gases**; Alzheimer = **donepezil, galantamine, rivastigmine**
- **Muscarinic blockers:** nonselective = **atropine, scopolamine, ipratropium, tiotropium**; M3 = **dicyclomine, oxybutynin**; M1 = **pirenzepine, telenzepine**
- **Nicotinic blockers:** ganglionic = **hexamethonium**; neuromuscular = **tubocurarine, pancuronium, atracurium, vecuronium**

ADRENERGIC PHARMACOLOGY

- Catecholamines: **norepinephrine (NE), epinephrine (EPI), dopamine (DA)**
- **NE** = main transmitter of **sympathetic postganglionic** fibers (exception: **eccrine sweat glands use ACh**)
- Synthesis: **tyrosine → L-DOPA → dopamine → NE → epinephrine**
  - Enzymes: **tyrosine hydroxylase** (rate-limiting) → AAAD → dopamine β-hydroxylase → **PNMT** (adrenal medulla)

CATECHOLAMINE LIFE CYCLE (AND WHAT BLOCKS EACH STEP)

- **Synthesis** – **tyrosine hydroxylase** (rate-limiting) → inhibited by **metyrosine**
- **Storage** – vesicles via **VMAT** → inhibited by **reserpine**
- **Release** – inhibited by **guanethidine**; **activated** (released) by **amphetamines, ephedrine, tyramine**
- **Termination** – mainly **reuptake** via **NET** (and DAT) → inhibited by **cocaine** and **tricyclic antidepressants (TCAs)**; then metabolized by COMT and MAO

Note: Memory tip: "**Met Reserve Guards at Coco's**": **Met**yrosine (synthesis), **Reserp**ine (storage), **Guan**ethidine (release), **Coc**aine/TCA (reuptake), in the same order as the life cycle.

- **Side by side:** synthesis – hemicholinium vs **metyrosine**; storage – vesamicol vs **reserpine**; release – botulinum vs **guanethidine**; termination – AChE inhibitors vs **cocaine/TCA** (reuptake)
- **Tyramine** is in aged foods (cheese, cured meats, beer, soy sauce, overripe fruit) and releases NE (migraine trigger)

ADRENERGIC RECEPTORS (SLIDES)

- **α1 – CONTRACTION**: vasculature, sphincters (vasoconstriction)
- **α2 – RELAXATION**: vessels, bladder and uterine walls (also presynaptic brake on NE release)
- **β1 – ONE HEART**: contractility, heart rate
- **β2 – TWO LUNGS**: bronchodilation

Note: Memory tip: 1 heart (β1), 2 lungs (β2). α1 = **squeeze**, α2 = **slow down**.

SYMPATHOMIMETICS (ADRENERGIC AGONISTS)

NONSELECTIVE

- **Norepinephrine** – **α1 > β1 >>> β2**; ↑ peripheral resistance; **first line in septic shock**
- **Epinephrine (adrenaline)** – **β1 = β2 > α**; **anaphylaxis**, shock (EpiPen)
- **Dopamine** – dose-dependent: **low → D1** (renal vasodilation); **mid → β1** (tachycardia); **high → α1** (vasoconstriction); first line in **septic shock with renal shutdown**

Note: Memory tip: Norepi = "**no** β2"; **Epi** = "**every**one" (β1 = β2), so it opens the airways in **anaphylaxis**. Dopamine climbs **D → β → α** as the dose rises.

ALPHA AGONISTS

- **α1 – phenylephrine, pseudoephedrine, oxymetazoline, tetrahydrozoline → decongestant** (vasoconstricts nasal vessels)
- **α2 – clonidine → hypertension**; off-label **ADHD, Tourette**
- **α2 – methyldopa → gestational hypertension, preeclampsia**

Note: Memory tip: α1 drugs are the **nose and eye drops** ("-zoline", "-ephrine"). **M**ethyldopa = **M**om's drug (safe in pregnancy).

BETA AGONISTS

- **Nonselective – isoproterenol** (β1 = β2 = β3) → **asthma**
- **β1 – dobutamine → cardiogenic shock, cardiac stress testing**
- **β2 – salbutamol (albuterol), terbutaline, isoxsuprine → asthma attacks, tocolysis** (stops preterm labor); SE: **tachycardia, tremors, nervousness, restlessness**

Note: Memory tip: DOBUtamine **DOBU**les heart pumping (β1 = heart). β2 agonists relax **smooth muscle** in the **lungs and uterus**.

SYMPATHOLYTICS (ADRENERGIC BLOCKERS)

ALPHA BLOCKERS

- **Nonselective:**
  - **Phenoxybenzamine** – **irreversible**
  - **Phentolamine** – **reversible**
  - Use: **pheochromocytoma** (before surgery)
- **α1-selective (-ZOSIN): prazosin, doxazosin, tamsulosin → benign prostatic hyperplasia**

(BPH); SE: **orthostatic hypotension**; **tamsulosin is most selective for the prostate**

- **α2 – yohimbine** (obsolete) → orthostatic hypotension, male erectile dysfunction

Note: Memory tip: PHEochromocytoma → **PHE**noxybenzamine and **PHE**ntolamine. Pheo triad (slides): **P**alpitations, **H**eadache, **E**xcessive sweating. **-zosin = "pee-zosin"**: helps **pee** (BPH).

BETA BLOCKERS (-OLOL)

- **Nonselective** (β1 and β2): **propranolol, pindolol, timolol, labetalol, carvedilol**
  - Uses: **post-MI, hypertension, heart failure**; also **hyperthyroidism**, **migraine prophylaxis (propranolol)**
- **β1-selective (cardioselective): atenolol, acebutolol, bisoprolol, esmolol, metoprolol, nebivolol** (and betaxolol)
  - **Esmolol – shortest-acting, given IV**

Note: Slide mnemonic: cardioselective β1 blockers = "**MANBABE**": **M**etoprolol, **A**tenolol, **N**ebivolol, **B**isoprolol, **A**cebutolol, **B**etaxolol, **E**smolol.

Note: Slide "**LOL**" trick: names starting **A–N** = **β1-selective**; **O–Z** = **nonselective** (propranolol, timolol); **not ending in "-olol"** (labeta**lol**, carvedi**lol**) = **alpha AND beta** blocker.

ADRENERGIC SUMMARY TABLE (FILL-IN FROM THE SLIDES, COMPLETED)

- **Agonists:** nonselective = **NE, epinephrine, dopamine**; α1 = **phenylephrine, pseudoephedrine, oxymetazoline, tetrahydrozoline**; α2 = **clonidine, methyldopa**; β nonselective = **isoproterenol**; β1 = **dobutamine**; β2 = **salbutamol, terbutaline, isoxsuprine**
- **Blockers:** nonselective α = **phenoxybenzamine, phentolamine**; α1 = **prazosin, doxazosin, tamsulosin**; α2 = **yohimbine**; nonselective β = **propranolol, pindolol, timolol, labetalol, carvedilol**; β1 = **atenolol, acebutolol, bisoprolol, esmolol, metoprolol, nebivolol**

——— END OF LESSON ———

LESSON 5: HORMONES AND DIABETES MELLITUS

Pancreas, insulin, and antidiabetic drugs by mechanism

PHARMACOLOGY | MIDTERM: TOXICITY, METABOLISM, AUTONOMICS, ANTIDIABETICS

How do I keep all the antidiabetic drugs straight?

Sort them by **what they do to glucose**: **boost insulin release** (secretagogues, incretins), **make tissues respond** (sensitizers), **dump sugar in urine** (SGLT2), **delay sugar absorption** (alpha-glucosidase inhibitors), or **replace insulin**. Then use the **name endings**, which almost always reveal the class.

ENDOCRINE BASICS

- **Exocrine** glands secrete into **ducts**; **endocrine** glands secrete **hormones into the blood**
- Endocrine = the **2nd major controlling system**; hormones act by **circulating, paracrine** (nearby cells) or **autocrine** (same cell)
- **Lipid-soluble** hormones (steroids, thyroid) enter the cell → nuclear receptor → new protein; **water-soluble** hormones (amines, peptides) bind a **surface receptor** → cascade
- Regulation by **negative feedback** (e.g. hypothalamus TRH → pituitary TSH → thyroid T3/T4, which inhibits both)

THE PANCREAS

- **Both exocrine and endocrine**; **99% acini** (digestive enzymes); the **islets of Langerhans** are scattered among them
- **Alpha (A) cells – 17% – glucagon**
- **Beta (B) cells – 70% – insulin**
- **Delta (D) cells – 7% – somatostatin**
- **F cells – pancreatic polypeptide**

Note: Memory tip: Beta = **B**iggest share and makes insulin ("**B** for **B**ring sugar into cells"). **A**lpha = **A**ntagonist (glucagon raises sugar).

- **Hormone of the fasting state: glucagon** (raises glucose by glycogenolysis and gluconeogenesis)
- **Counterregulatory hormones** (oppose insulin): **glucagon, growth hormone, cortisol, epinephrine**

Note: Memory tip: the counterregulatory four are the **stress and fasting** hormones (**G**lucagon, **G**rowth hormone, **C**ortisol, **E**pinephrine): they push sugar **up**, while insulin alone pushes it **down**.

INSULIN

- Small protein: **A and B chains** joined by **disulfide bridges**; **proinsulin** includes the **C-peptide** (marker of the body's own insulin)
- **Release:** glucose enters β-cell via **GLUT2** → metabolism → ↑ **ATP** → **closes K+ (ATP) channel** → **depolarization** → **Ca2+ enters** → **exocytosis** of insulin
  - **Sulfonylureas** block this K+ channel → insulin release
- **Effects:** binds receptor → **GLUT-4** moves to the membrane → glucose enters (muscle, fat)
  - **Liver:** ↓ glycogenolysis and gluconeogenesis; ↑ glycogen synthesis, lipogenesis
  - **Muscle:** ↑ amino acid uptake, protein synthesis, glycogen synthesis
  - **Adipose:** ↑ triglyceride storage; ↓ lipolysis

DIABETES MELLITUS

- **Elevated blood glucose** from absent or inadequate insulin secretion, with or without impaired insulin action
- Classic **3 Ps**: **polyphagia** (appetite), **polydipsia** (thirst), **polyuria** (urination)
- **Type 1: β-cell destruction** (usually **autoimmune**); absolute insulin deficiency; mostly **< 30 years old**; **insulin-dependent**;

sudden onset, thin, **ketoacidosis common**, autoantibodies present, twin concordance **50%**

- **Type 2: insulin resistance + relative insulin deficiency**; may start with **diet, exercise, oral drugs**; gradual, adults, often obese, ketoacidosis rare, twin concordance **90%**, **90–95%** of diabetics
- **Gestational DM** – risks: pre-eclampsia, macrosomia, caesarean delivery, later T2DM
- **Metabolic syndrome** – any **3 of 5**: waist ≥102 cm (men) / ≥88 cm (women); triglycerides ≥150 mg/dL; HDL <40 (men) / <50 (women); BP ≥130/85; fasting glucose ≥100 mg/dL

DIAGNOSIS

- **Random glucose ≥ 200 mg/dL (11.1 mM)** with symptoms, **or**
- **Fasting glucose ≥ 126 mg/dL (7.0 mM)**, **or**
- **2-hour OGTT glucose ≥ 200 mg/dL**, **or**
- **HbA1c ≥ 6.5%**
- **HbA1c reflects glucose control over the past ~3 months** (lifespan of red blood cells)

Note: Memory tip: "**126 fasting, 200 random, 6.5 A1c**": the three numbers to know cold.

INSULIN PREPARATIONS (BY SPEED)

- **Rapid-acting – lispro, aspart, glulisine** – onset 5–15 min, peak 1–1.5 h, lasts 3–4 h
- **Short-acting – regular insulin** – onset 30–60 min, peak 2 h, 6–8 h; the one used **IV in emergencies**
- **Intermediate – NPH** (neutral protamine Hagedorn) – onset 2–4 h, peak 6–7 h, 10–20 h
- **Long-acting – glargine, detemir, degludec** – **flat** (no peak); glargine ~24 h, detemir 17 h, **degludec > 42 h**
- **Most common side effect of insulin: hypoglycemia** (shaking, fast heart rate, hunger, sweating, confusion, dizziness)

Note: Memory tip: Rapid = "**LAG**" (**L**ispro, **A**spart, **G**lulisine): a lag that's actually fast. Long = "**GDD**" (**G**largine, **D**etemir, **D**egludec); **De**gludec lasts **De**cades (the longest, > 42 h). **R**egular is the only one that goes **R**ight into the vein.

ORAL AND NON-INSULIN DRUGS, BY MECHANISM

1. INSULIN SECRETAGOGUES (SQUEEZE MORE INSULIN OUT)

- **Sulfonylureas** – **close β-cell K+ channels** (block K+ efflux) → **more insulin release**
  - **1st generation** (no longer used): **tolbutamide, chlorpropamide, tolazamide**
  - **2nd generation** (more potent): **glyburide, glipizide, glimepiride, gliclazide**
  - **Major side effect: hypoglycemia** (e.g. **gliclazide**); also **weight gain**
- **Meglitinides** – same site as sulfonylureas but **faster onset, shorter duration**; taken **before meals** for **postprandial** glucose: **repaglinide, mitiglinide**

Note: Memory tip: sulfonylureas start with "**gli-**" and end in "**-ide**"; meglitinides end in "**-glinide**" (re**paglinide**). Both **squeeze** the β-cell, so both cause **hypoglycemia and weight gain**.

2. INSULIN SENSITIZERS (MAKE INSULIN WORK BETTER)

- **Biguanide – metformin**
  - MOA: **activates AMPK** → ↓ **hepatic glucose production**
  - **First-line in type 2 DM**; also used in **PCOS**
  - SE: **GI** (most common: anorexia, nausea, vomiting, diarrhea), **vitamin B12 deficiency**, **lactic acidosis**; **weight loss / neutral**, **no hypoglycemia alone**
- **Thiazolidinediones (-GLITAZONE) – pioglitazone, rosiglitazone**
  - MOA: ligands of **PPAR-gamma** (nuclear receptor) → better insulin response in fat and muscle

- SE: **fluid retention (heart failure)**, **bone loss (fractures)**, **anemia**, **weight gain and edema**

Note: Memory tip: Metformin SEs = "**GLB**": **G**I upset, **L**actic acidosis, **B**12 deficiency. TZD SEs = "**FAB-W**": **F**luid/heart failure, **A**nemia, **B**one fractures, **W**eight gain. "**-glitazone**" = "**glit**ter in the **zone**": sensitizes the tissues.

3. INCRETIN MODULATORS (BOOST THE GUT HORMONE)

- **Incretins (GLP-1, GIP)** → stimulate insulin release and **inhibit glucagon** after an oral glucose load; **DPP-4** inactivates them
- **GLP-1 receptor agonists – exenatide, semaglutide, liraglutide, albiglutide, dulaglutide**
  - **Weight loss**; SE: **GI disturbances**, **acute pancreatitis**
- **DPP-4 inhibitors (-GLIPTIN) – sitagliptin, saxagliptin, linagliptin, alogliptin, vildagliptin**
  - Keep incretins around longer; **weight neutral**; SE: **headache, nasopharyngitis, URTI**

Note: Memory tip: GLP-1 agonists end in "**-glutide**" or "**-natide**" (inject the hormone itself, weight **loss**). DPP-4 inhibitors end in "**-gliptin**" (**gl-i-ptin**: they **grip** the enzyme). Both work only when there's sugar to respond to.

4. EXCRETERS (SGLT2 INHIBITORS, -GLIFLOZIN)

- **Canagliflozin, dapagliflozin, empagliflozin**
- Block **SGLT2** (sodium-glucose transporter) in the proximal tubule → **glycosuria** → lower blood glucose; **weight loss**
- SE: **genital and urinary tract infections**

Note: Memory tip: "-gli**FLOZ**in" = glucose **FLOWS** out in the urine, and sugary urine feeds **infections**.

5. ALPHA-GLUCOSIDASE INHIBITORS (DELAY CARB ABSORPTION)

- **Acarbose, miglitol**
- Inhibit intestinal **alpha-glucosidase** → slower digestion of starch and disaccharides → ↓ **post-meal glucose spikes**; **no effect on fasting glucose**
- SE: **flatulence, diarrhea, abdominal pain** (undigested carbs reach the colon)

Note: Memory tip: A-CARB-ose blocks **carbs**; unabsorbed carbs get fermented, so **gas**.

EFFECT ON WEIGHT (COMMON EXAM QUESTION)

- **Weight gain:** insulin, meglitinides, sulfonylureas, thiazolidinediones
- **Weight neutral:** alpha-glucosidase inhibitors, DPP-4 inhibitors
- **Weight loss:** GLP-1 agonists, metformin, SGLT2 inhibitors, pramlintide

Note: Memory tip: drugs that **add insulin** (injected or squeezed out) or **store fat** (TZDs) **add weight**; drugs that **remove sugar** (urine, less liver output) or **curb appetite** (GLP-1) **remove weight**.

MANAGEMENT

- Four parts: **diet, education, glycemic targets, treatment options**
- **Diet** – balanced, limit carbohydrates, use **monounsaturated fats** (olive, canola, nuts, avocado); **Mediterranean-style**
- **Education** – type of diabetes, **self-monitoring**, signs of **hypoglycemia** (patient and family)
- **Glycemic targets:**
  - **HbA1c < 7%** (**< 8%** for frail elderly)
  - **Pre-meal glucose 90–130 mg/dL** (5–7.2 mmol/L)
  - **1-h postprandial < 180 mg/dL**; **2-h postprandial < 150 mg/dL**
  - **BP target < 140/90**
- **Type 1 treatment – 3–4 insulin injections a day**; **rapid + long-acting** analogs to mimic normal secretion; **insulin units = weight (kg) × 0.55**

- **Type 2 treatment – caloric restriction, more exercise**, then **metformin first**; add oral agents or insulin based on HbA1c

COMPLICATIONS

- **Microvascular: retinopathy** (aneurysms, hemorrhages, cotton-wool spots), **nephropathy**, **neuropathy** (foot ulcers)
- **Macrovascular: cerebrovascular disease (stroke)**, **cardiovascular disease (heart attack)**, **peripheral vascular disease**

Note: Memory tip: micro = the "**-opathies**" of small vessels: **eyes, kidneys, nerves**. Macro = big arteries: **brain, heart, legs**.

——— END OF LESSON ———